Chapter 1 - Introduction

Lecture Mobile Communication

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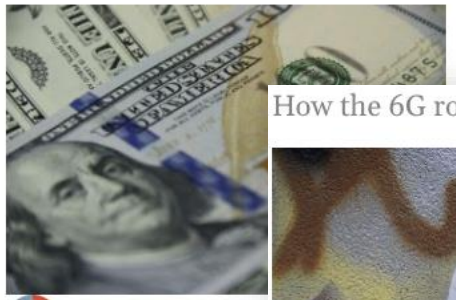
Class objectives

The other day on telecomstechnews.com...

Karlsruhe Institute of Technology

GSMA: North America mobile industry fuels $1.6T economic growth



A new report from the GSMA has revealed the immense economic growth power of the mobile industry across North America, with its contribution nearing $1.6 trillion in 2024. This figure

How the 6G rollout will learn from 5G's mistakes

The telecoms industry, learning from 5G's mistakes, is already laying a pragmatic and cost-conscious foundation for the 6G rollout.

A new report from ABI Research and InterDigital spotlights The 3rd Generation Partnership

Urgent digital migration could save UK services billions

Migrating the UK's critical national infrastructure (CNI) from ageing analogue networks to modern digital services could generate a net economic benefit of over £3 billion by 2040.

A study – conducted by Assembly Research and released today by BT – shows the immense

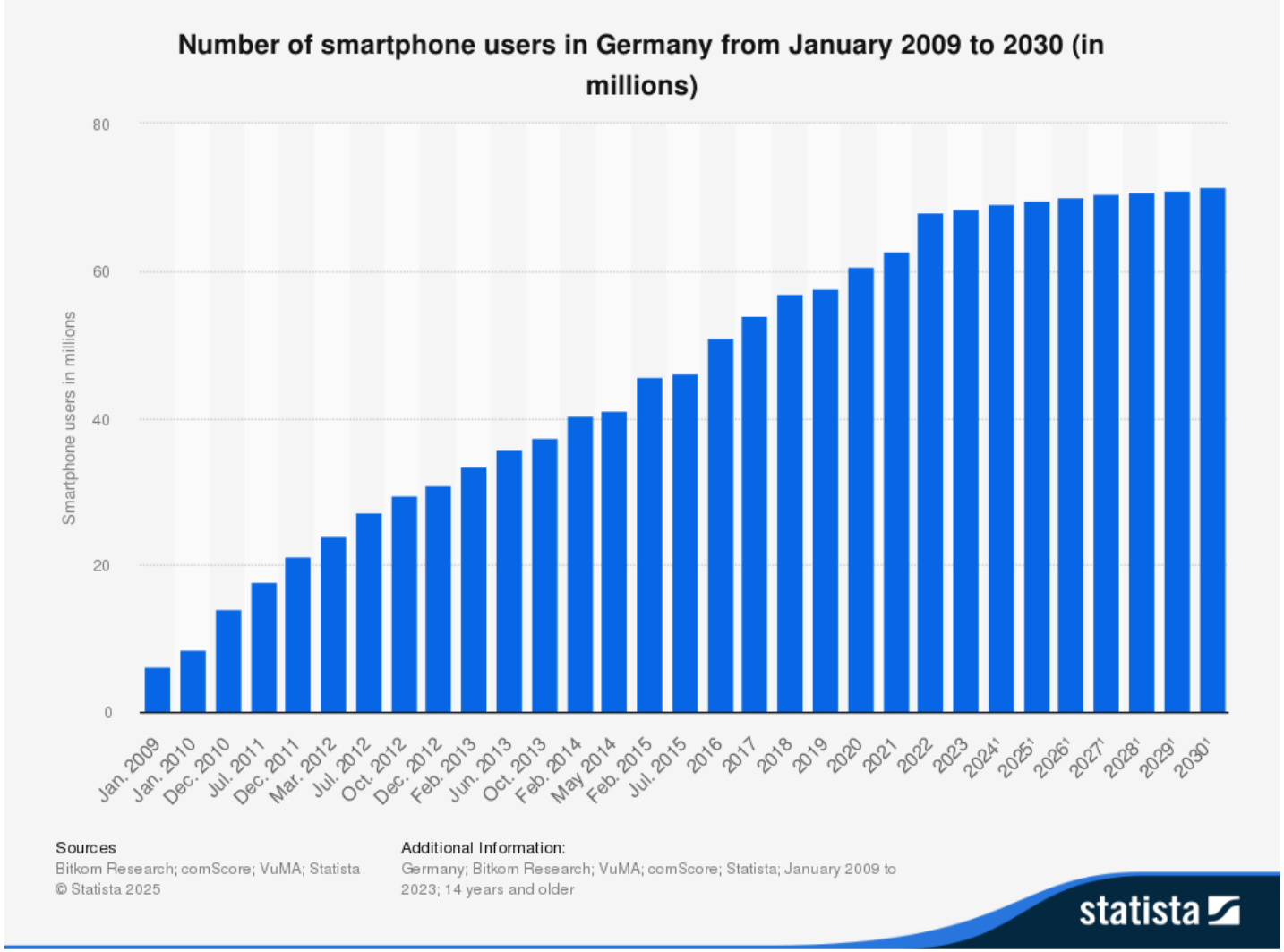
On-premise edge and private 5G key to industrial AI and security

Investing in industrial tech pays off fast, with 87% of companies adopting on-premise edge and private 5G networks seeing a ROI (Return on Investment) in one year.

New research – a third joint effort between Nokia and GlobalData – makes it clear that private wireless and edge computing are now the go-to tools for building the smart, secure, and efficient factories and facilities of the future.

The '2025 Industrial Digitalization Report' gathered insights from 115...

Triumph of the cell phone



Other observations

- Significantly more SIM cards than landlines
  - 110 million vs. 38 million → almost a factor of 3!
- Number of mobile devices with Internet connection greater than that of stationary devices
  - Notebooks, smartphones, etc. enable Internet access anywhere
- Two important (but different) challenges
  - Wireless Internet access: Communication via wireless (radio) connection
  - Mobile Internet access: Communication from changing access point

Objectives and topics of the lecture

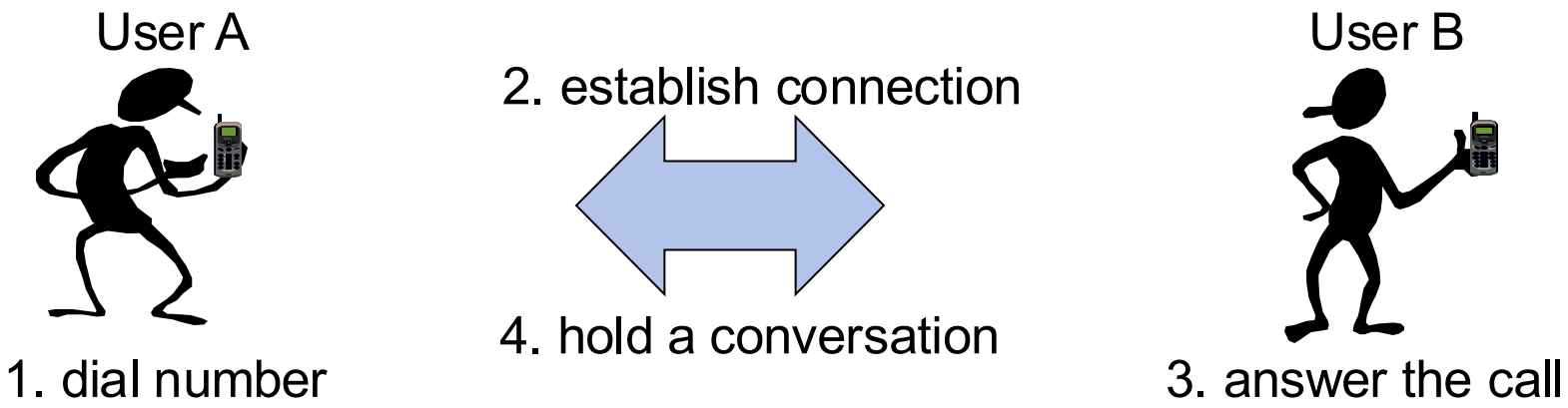
- Contents
  - Important issues in mobile communication systems
  - "Fixed network counterparts"???
  - Examples
    - GSM - ISDN
    - WLAN - Ethernet
    - Mobile IP - IP

- What is different?
  - What new problems/tasks arise?
    - How are these solved?
  - Basic principles of mobile communication systems
    - Using the example of protocols and architectures

Example: GSM - ISDN

- ISDN: Standard for the digital fixed network
- GSM: Digital mobile communications standard

- ... nothing really changes for users in terms of operation
  - Advantage: accessible everywhere
  - Disadvantage: Poorer voice quality



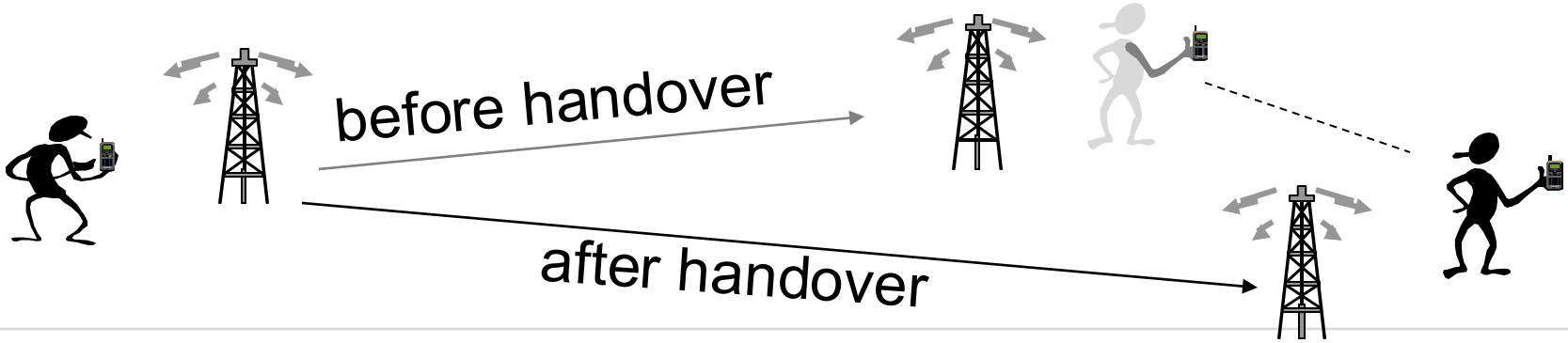
- What will change for the network?

Example: GSM - ISDN

- New problem: position management
  - ISDN
    - Locations of user A and B are known
      - Phone number specifies location
  - GSM
    - Users A and B can be mobile
      - Phone number does not allow conclusions about location
    - Questions
      - How is a mobile device located worldwide?
      - Where should the network establish the connection to?
    - GSM requires a location service
      - Assignment of telephone number to current location

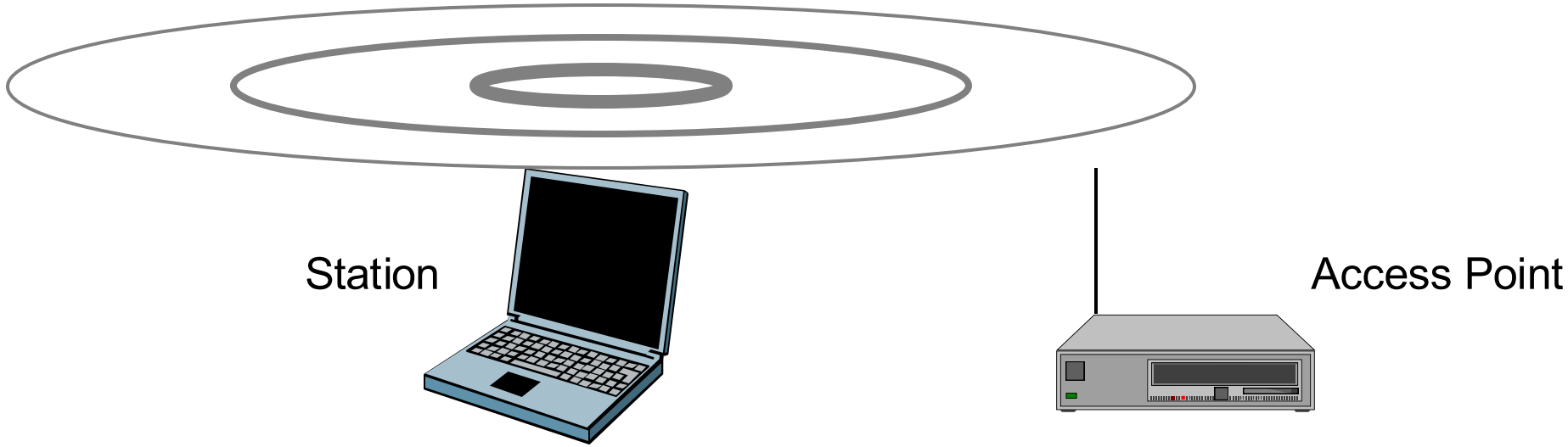
Example: GSM - ISDN

- New problem: Handover
  - ISDN
    - A user's location does not change during a connection
      - Access point to the network (NTBA) remains the same
  - GSM
    - Users can be mobile during a connection
      - User may have to change access point (base station)
    - What happens to the already established connection?
    - GSM must support handover
      - "Bending" the connection when changing the access point



Example: WLAN - Ethernet

- Ethernet (IEEE 802.3)
  - Standard for wired local networks
- WLAN (IEEE 802.11, „WiFi“)
  - Standard for wireless local area networks



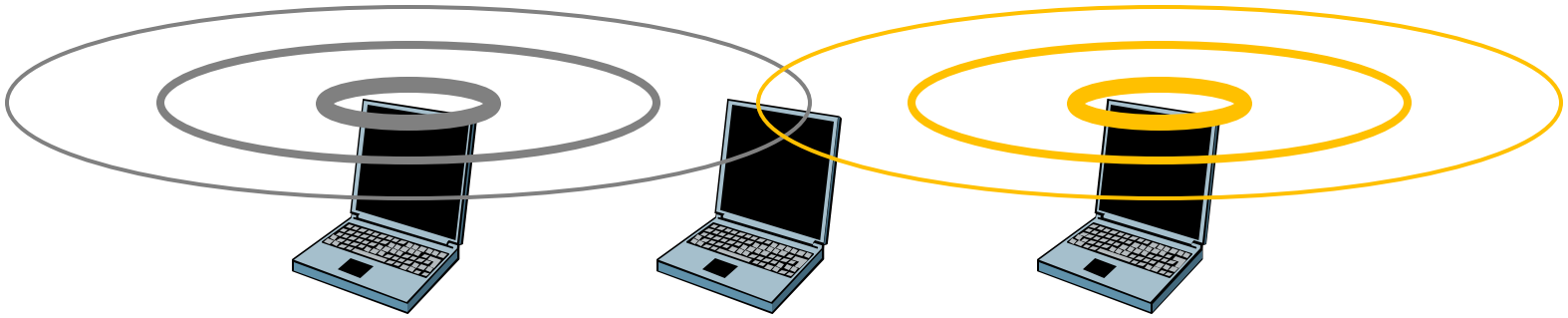
Fundamental differences between the two, or ... simply a change of transmission medium?

Example: WLAN - Ethernet

- New problem: Overlapping LANs
  - 802.3 (Ethernet)
    - No overlapping of LANs possible
      - Network card is connected to one LAN via one cable
      - Station only receives data units of a LAN
      - Coupling several LANs via bridges or routers
  - 802.11 (WLAN)
    - Overlapping of LANs possible
      - Station receives data units from all stations within range
      - Stations within range can belong to other LANs?
    - Assignment of data units to a specific LAN?
      - To which LAN does a broadcast data unit belong?
    - Indication of LAN affiliation in each data unit

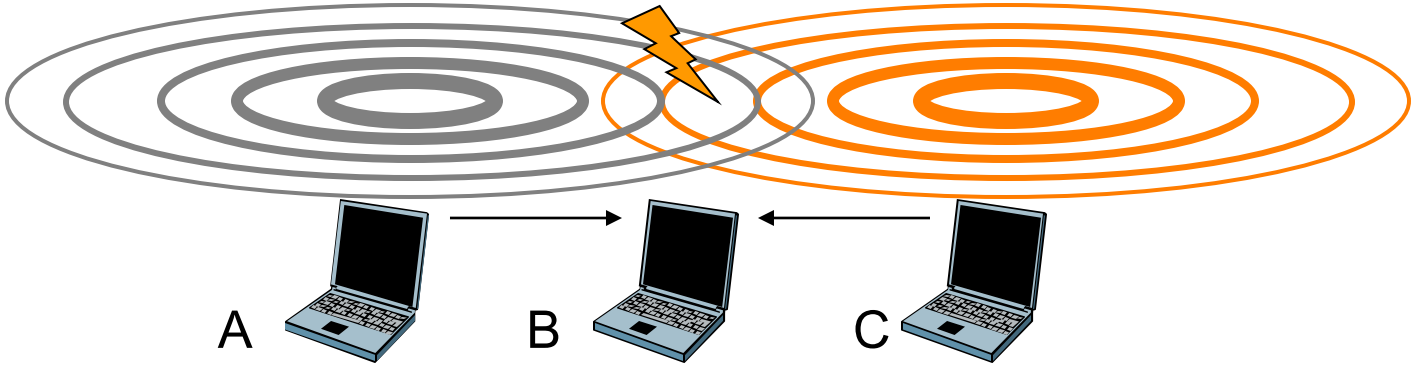
Example: WLAN - Ethernet

- New problem: semi-broadcast medium
  - 802.3 (Ethernet)
    - Stations are connected to each other via broadcast medium
      - Stations receives data units of all stations in the network
      - Collision detection by transmitter possible
  - 802.11 (WLAN)
    - Stations are connected via a semi-broadcast medium
      - Stations receives data units only from stations within range
      - Transmitter cannot detect collision
      - Hidden terminal problem
    - Media access mechanisms must be adapted



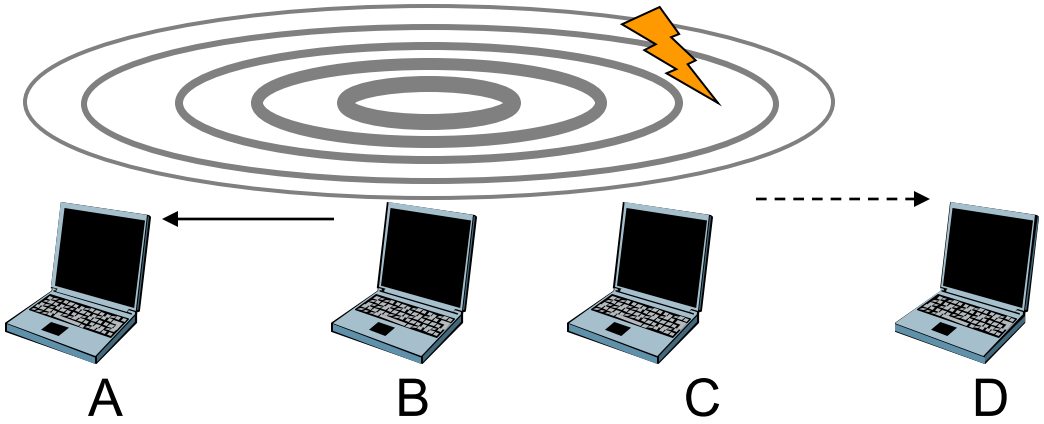
Hidden terminal problem

- Example situation
  - Station A transmits to station B
  - Station C cannot detect signals from station A
  - Station C wants to transmit to station B
    - Medium is free for station C (carrier sense fails)
  - Collision at station B
    - Station A cannot detect this (collision detection fails)
  - Station A is "hidden" for station C



Exposed terminal problem

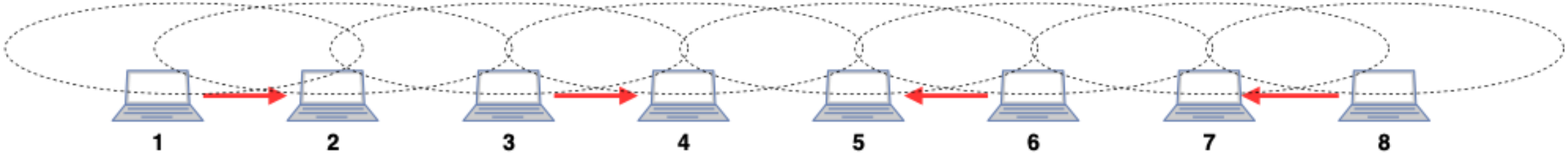
- Example situation
  - Station B transmits to station A
  - Station C wants to transmit to station D
  - Station C must wait
    - Carrier Sense signals a "busy" medium
  - However, as station A is outside the range of station C, this is unnecessary
  - Station C is “exposed" to Station B



Exercise: Hidden and Exposed Terminals

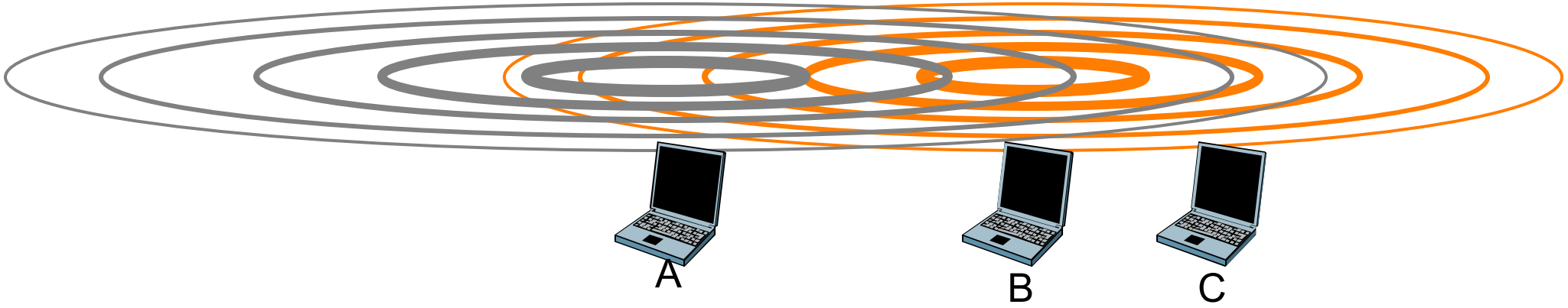
We assume that eight mobile stations, numbered 1 to 8, are arranged in a line such that each station's reception range only reaches its immediate neighbors. We further assume that communication occurs according to the pattern 1→2, 3→4, 6→5, 8→7.

- Are there any hidden or exposed terminals? If so, which ones?
- Are any messages lost due to collisions?



Near and far end devices

- Example situation: Stations A and B transmit, station C is intended receiver
  - Signal strength decreases quadratically with distance
  - Signal from Station B "drowns out" the signal from Station A
  - Station C cannot hear station A



- Problem must be taken into account when designing a wireless network
  - Depending on the type of mechanism used, precise power control is required to avoid "drowning out".

Example: WLAN - Ethernet

- New problem: interception of the radio medium
  - 802.3 (Ethernet)
    - Stations are connected to each other via a wired medium
      - Physical access to the network required for eavesdropping
  - 802.11 (WLAN)
    - Stations are connected to each other via a wireless medium (radio)
      - Network can still be tapped over a certain distance
    - How can eavesdropping be made more difficult or prevented?
    - Use of security mechanisms in the data link layer

Summary

- Questions from the following areas
  - Physical layer, e.g.
    - Signal propagation
    - Modulation techniques
    - Space, frequency, time and code multiplexing
  - Data link layer, e.g.
    - (Slotted) Aloha
    - Carrier Sense Multiple Access
  - Network layer and network architecture
    - Call switching in mobile networks
    - Mobile Internet Protocol (MobileIP)
  - *Transport layer and above*
    - *E.g. Mobile TCP*
  - *Security in mobile communication systems*
    - *E.g. subscriber authentication in mobile networks*

Contents

WLAN, Bluetooth

GSM, UMTS, LTE, 5G

Mobility management

Media access

Wireless transmission

More symbols

- Additional information - not relevant to the exam

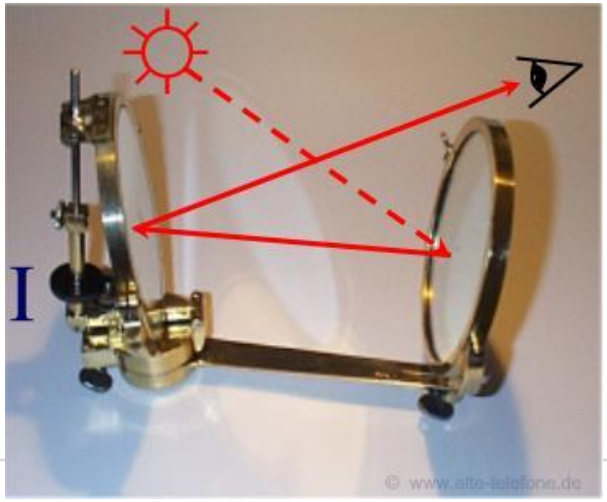
- Updated slides

A brief history of mobile communication

About the history

- Wireless communication is "ancient"
  - ... well, technology has made significant progress
- Heliograph (405 BC, Greece)
  - Patterns of light pulses using mirrors
  - Used for military purposes from 1875
    - Replaced by electric telegraph due to high weather dependency
  - Schematic representation

Button for tilting the mirror



[1.9]

Torch Telegraph

"Since ancient times, fire has been one of the classic media of signaling. Usually the signal with the help of burning torches, which were placed between the battlements the battlements of a signal tower were placed between the battlements of a signal tower."

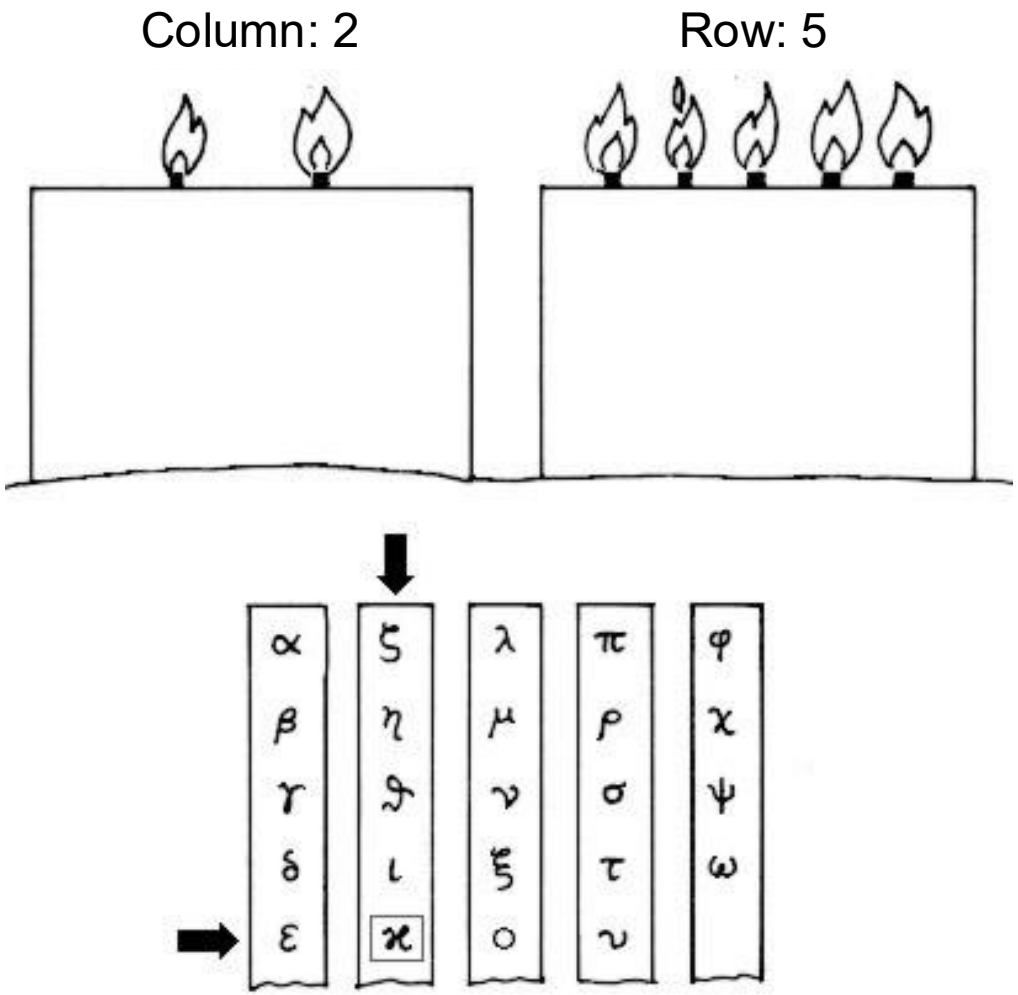
[1.3]

Roman Limes watchtower with torch signal. Copperplate engraving, 1785

Watercolor by A. Roloff

Torch Telegraph

- Suggested by Polybius in antiquity
  - 150 BC, Greece
- Functionality
  - Each letter of the alphabet (user data) is assigned a torch code (symbol)
    - 5 torches - each determining the row and the column
- Problems
  - Short range
  - Operators must be able to read and write
- No evidence of historical use



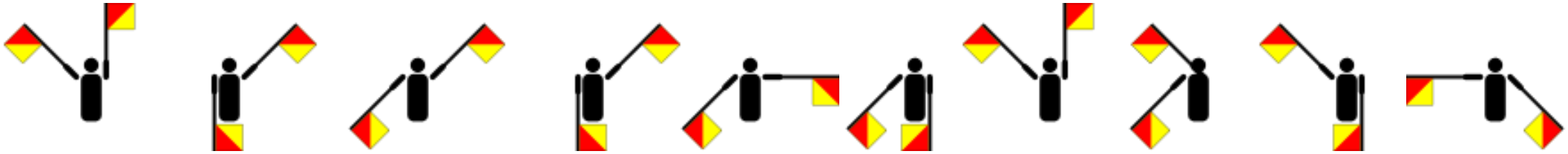
[1.10, 1.3]

Exercise: Torch Telegraph

- Approximately 8 characters per minute can be transmitted. What data rate can be achieved?

Flag telegraph

- 1794: Claude Chappe (1763 - 1805)
  - Flags ("semaphores"), pointers
    - Positions of flags encoded characters
  - Example



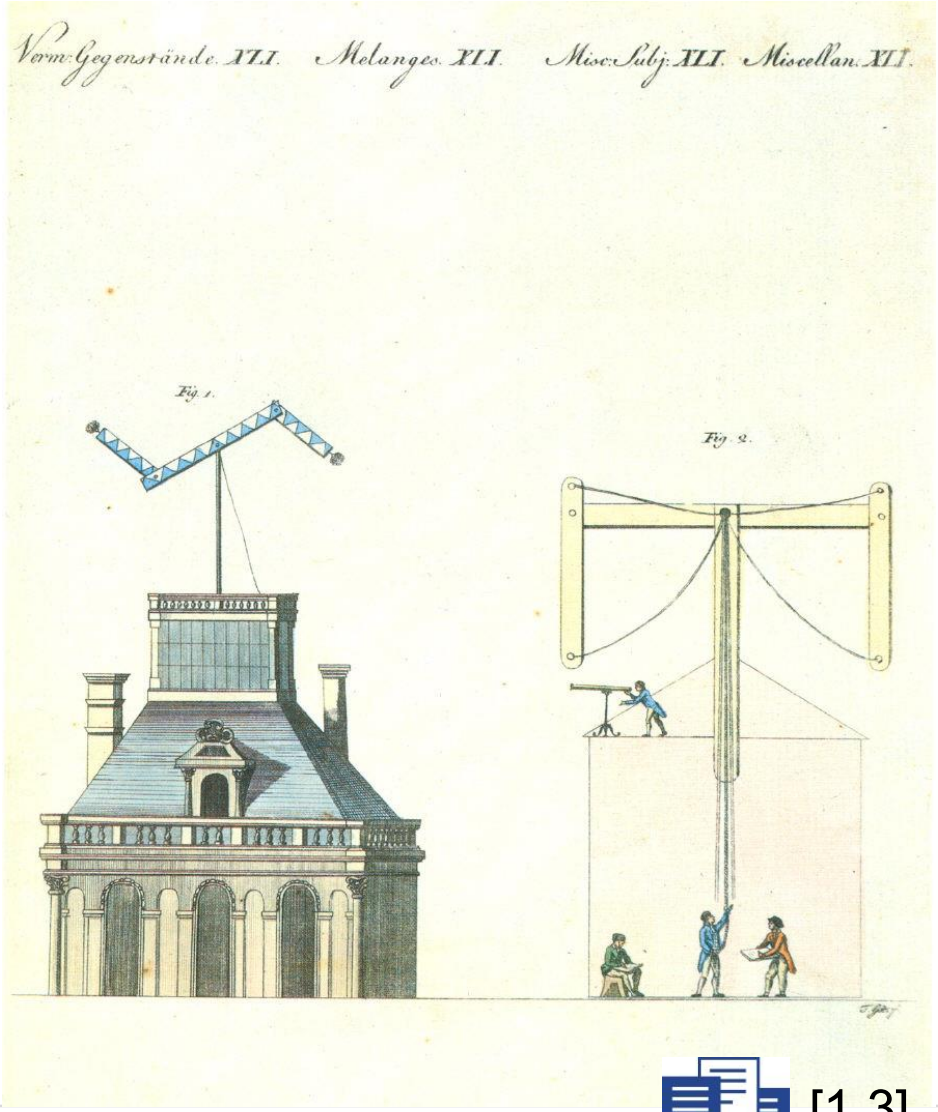
T E L E M A T I C S

- First telegraph line (1794) from Paris to Lille
  - Approx. 210 km with 23 intermediate stops
  - Transmission of a long message from Lille to Paris within an hour
  - Delivery by mounted messenger would have taken 24 hours
  - ... strategic advantage in the event of a feared attack on England
    - Lille is located at the narrowest point of the English Channel

[1.11]

Flag telegraph

- Functionality of a flag telegraph
  - 7 positions per pointer
  - 2 positions per crossbar
  - → 7 x 2 x 2 x 7 = 196 figures
  - 92 <u>easily recognizable</u> symbols linked with meaning via code directory
  - 2 <u>symbols</u> per <u>message unit</u>
    - Page (1...92)
    - Line (1...92)
    - Thus 92 x 92 = 8464 different message units



[1.3]

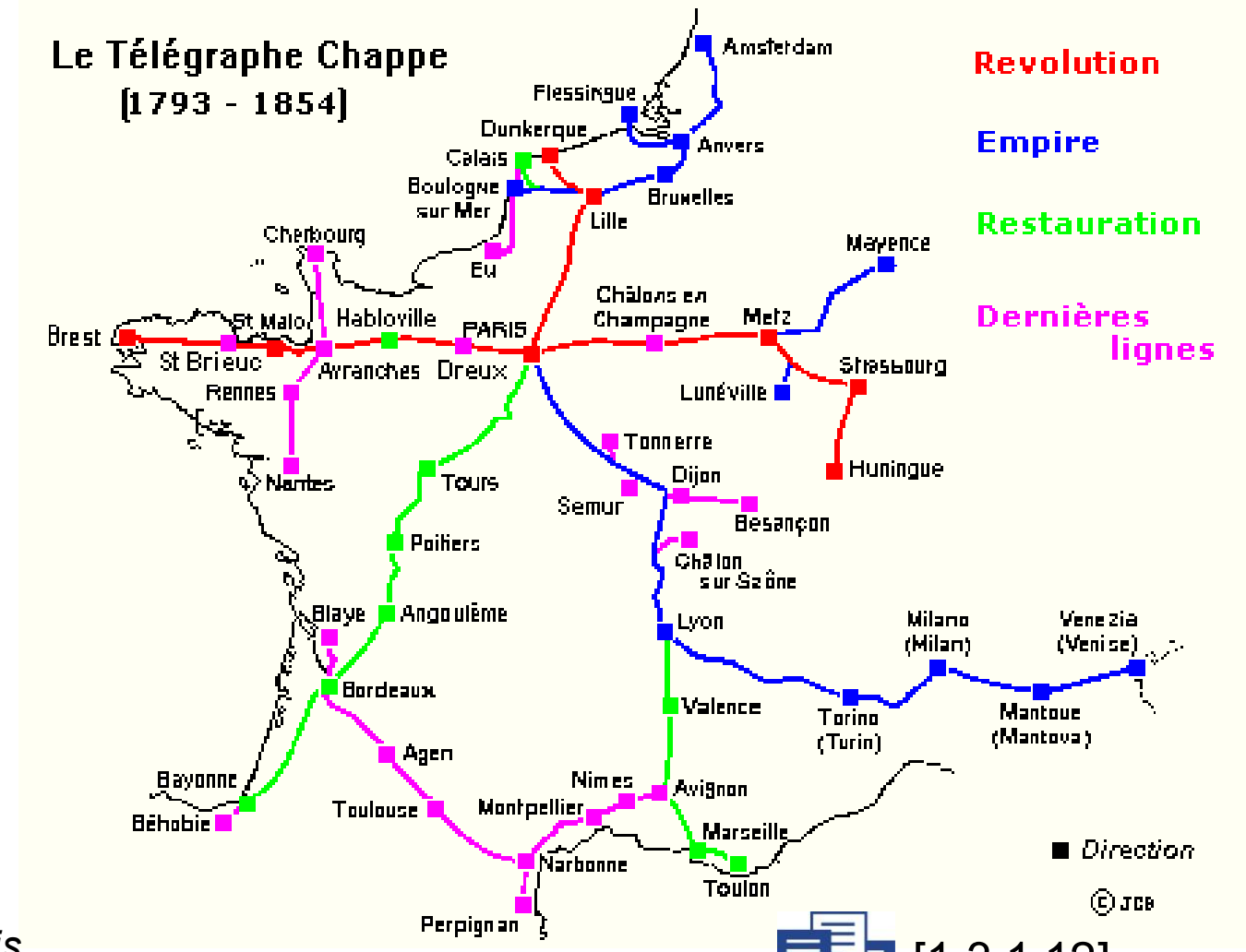
Flag telegraph - Network in France

Pleumeur-Bodou

Sainte Foy-lès-Lyon

Marcy sur Anse

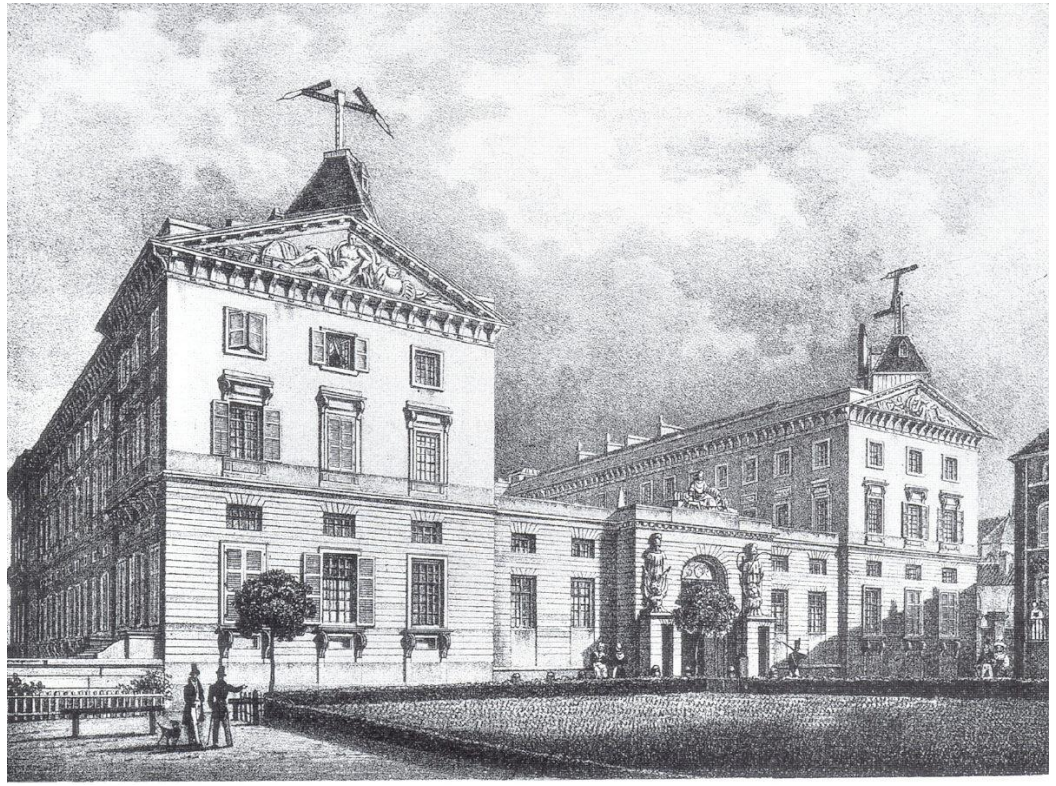
Church of St. Pierre on the Montmartre hill in Paris



[1.3,1.12]

Flag telegraph

- "Initially, Metz was a station on the Paris-Strasbourg line; in in 1813/1814 a branch was added, so that the Palace of Justice had two telegraph stations. two telegraph stations housed two telegraph stations."



[1.3]

Exercise: Flag telegraph

- Each symbol is displayed for approx. 20 seconds to avoid transmission errors. What data rate can be achieved?

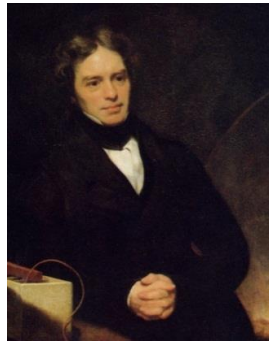
A telegraph for Karlsruhe

- "Carlsruher Zeitung" from October 20, 1794
  - First report on telegraphs in France
    - Great interest
      - Previously: maximum notification speed: gallop of a horse
    - Information is detached from the material!
      - Claude Chappe: "I write in space"
  - Böckmann from Karlsruhe makes new inventions in the field of telegraphs
    - Professor of Mathematics and Physics in Karlsruhe
    - ... something must be done
    - ... "not only because of the almost unbelievable transmission speeds" [1.3]
    - ... state of war - so nobody wanted to take over France's findings

[1.3]

Important personalities and developments.

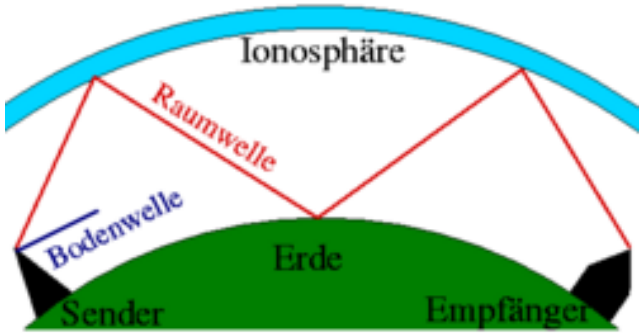
- The use of radio for communication is of particular interest for the lecture.
  The following researchers, among others, made special contributions to this course
  - Michael Faraday (1791 - 1867)
    - 1831: Demonstration of electromagnetic induction
  - James Clerk Maxwell (1831 - 1879)
    - 1864: Theory of electromagnetic fields
    - Wave equations
  - Heinrich Rudolf Hertz (1857 - 1894)
    - 1886: Experimentally demonstrates the wave character of electrical transmission through space
      - In Karlsruhe, on the site of today's Hertz auditorium (KIT)



Important personalities and developments.

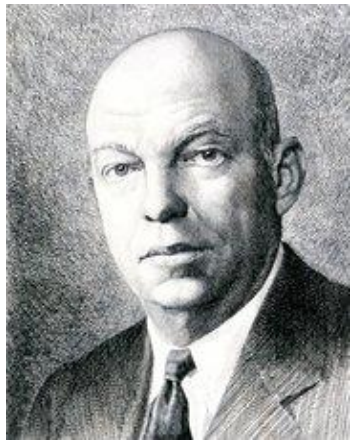
- 1896: Guglielmo Marchese Marconi (1874 - 1937)
  - First demonstration of wireless telegraphy (digital!)
  - Long-wave transmission, high transmission power (> 200kW)
  - Wireless transmission over 1.5 miles
  - 1897: Foundation of *Marconi's Wireless Telegraph Company Ltd*.
  - 1899: First wireless connection across the English Channel
  - 1901: First transatlantic radio transmission

- 1906: First conference of the International Telecommunication Union (ITU)
- 1907: Commercial transatlantic connections
  - Very large base stations (30 x 100m high antenna masts)
- 1915: Wireless voice transmission New York - San Francisco
- 1920: Discovery of short-wave by Marconi
  - 3 MHz - 30 MHz, wavelength from 100m to 10m
  - Reflection at the ionosphere (space wave)
  - Invention of the vacuum tube (1906, Lee DeForest and Robert von Lieben) Enables smaller transmitters and receivers



Important personalities and developments.

- 1926: First train telephone
  - Hamburg - Berlin route
  - Wires parallel to the railroad line
- 1928: Many field trials with TV
  - Color TV, News, Atlantic
- 1933 E. H. Armstrong (1890 - 1954)
  - Invention of frequency modulation



[1.1]

Historical mobile networks

- 1958 A-network in Germany
  - First test runs in 1953
    - Cell phone call from a VW Beetle
    - Mobile radio: weight 16 kg, price: 8000 DM
      - For comparison: VW Beetle price at the time: 5000 DM
  - analog, 160 MHz, connection setup only from the mobile station, no handover, 80% coverage, 11,000 subscribers (1971)
    - Manual mediation by "lady from the office"
    - 137 zones each 30 km in diameter
  - Switched off in 1977

[1.6]

Historical mobile networks

- 1972: B-network in Germany
  - Introduction to the 1972 Olympic Games in Munich
  - analog, 160 MHz, connection can also be established from the fixed network (but the location of the mobile station must be known), no handover
    - No more "lady from the office" required
    - Devices mainly used in cars
  - also in A, NL and LUX, 27000 participants in D (1979)
  - Switched off in 1994
- 1979: NMT (Nordic Mobile Telephone System), 450 MHz (Scandinavia)
  - Automatic switching centers
- 1982: Start of the GSM specification
  - Through "Group Special Mobile"
    - Should develop for CEPT standard
  - Goal: pan-European digital mobile network with roaming
- 1983: AMPS in America (Advanced Mobile Phone System)
  - analog
- 1984: CT-1 standard (Europe) for cordless telephones

Historical mobile networks

- 1986: C-network in Germany
  - analogue voice transmission, 450 MHz, handover possible, digital signaling, automatic localization of the mobile station
  - Services: FAX, modem, Datex-P, e-mail, 98% coverage
  - 12/2000 switched off
- 1991: Specification of the DECT standard
  - Digital European Cordless Telephone (today: Digital Enhanced Cordless Telecommunications)
  - 1880-1900 MHz, ~100-500 m range, 120 duplex channels, 1.2 Mbit/s data transmission, voice encryption, authentication, several 10,000 users/km^2 , use in 40 countries

(More) Current mobile networks

- 1992: Launch of GSM
  - in Germany as D1 and D2, fully digital, 900 MHz, 124 carrier frequencies
  - automatic localization, handover, cellular
  - Roaming in Europe - now also in over 150 countries worldwide
  - Services: Data with 9.6 kbit/s, FAX, voice ... SMS (!)
- 1994 E-network in Germany
  - GSM with 1800 MHz, smaller cells, currently 11 countries (DCS-1800)
  - than E-Plus in Germany
- 1996 HiperLAN (High Performance Radio Local Area Network)
  - ETSI (European Telecommunications Standards Institute)
  - Standardization of type 1: 5.15 - 5.30 GHz, 23.5 Mbit/s
  - Proposals for types 2 and 3 (both 5 GHz) and 4 (17 GHz) as wireless ATM extensions (up to 155 Mbit/s)
  - ... of no practical significance today, but has had an influence on further standardization
- 1997 Wireless LAN - IEEE 802.11
  - IEEE standard, 2.4 - 2.5 GHz and infrared, 2 Mbit/s
  - many proprietary products earlier

Current mobile networks

- 1998 Specification of GSM successors
  - UMTS (Universal Mobile Telecommunication System) as a European proposal for IMT-2000 (International Mobile Telecommunications-2000, adopted by the ITU)
- 1999
  - Other wireless LANs
    - IEEE standard 802.11b, 2.4 - 2.5 GHz, 11 Mbit/s
  - Bluetooth for personal area networks, 2.4 GHz, < 1 Mbit/s
  - Decision on IMT-2000
    - Several "family members": UMTS, CDMA2000, UWC-136
  - Starting WAP (Wireless Application Protocol)
    - Beginning of the Internet/mobile communications merger
    - Access to a wide range of information services via a cell phone

Current mobile networks

- 2000: GSM with higher transmission rates
  - HSCSD offers up to 57.6 kbit/s
  - First GPRS installations with up to 115.2 kbit/s
- 2002: First test installations of UMTS in Germany
- 2004: Commercial launch of UMTS in Germany
- 2006: Commercial launch of HSDPA in Germany
  - Higher data rates for UMTS
- 2010: Auction of new LTE frequencies (360 megahertz)
- 2011: Start of LTE expansion (nationwide availability since approx. 2015)
- 2014: Expansion LTE-Advanced
- 2019: Auction of new 5G frequencies
- 2020: Commercial launch of 5G

Comparison of mobile phone generations

Generation	Mobile technology	Maximum bandwidth
1G	AMPS	-
2G	GSM	9.6 kb/s
2.5G	HSCSD	57.6 kb/s
	GPRS	115 kb/s
2.75G	EDGE	236 kb/s
3G	UMTS	384 kb/s
3.5G	HSPA	14.4 Mb/s
	HSPA+	42.2 Mb/s
3.9G	LTE (up to Cat 4)	150 Mb/s
4G	LTE Advanced (LTE-A from Cat 6)	300 to 600 Mb/s
4.5G	LTE Advanced Pro (LTE-AP from Cat 11)	1 Gb/s
5G	5G	10 Gb/s

Source: https://www.elektronik-kompendium.de/sites/kom/0406221.htm

Exercises

1.1 What new tasks need to be solved in mobile communication compared to fixed network communication?

1.2 What is a handover?

1.3 Explain the problem of hidden and delivered end devices.

1.4 What are the consequences of the problem of near and far end devices?

1.5 Illustrate the difference between symbols and user data using the torch and flag telegraph.

References and further reading

[1.1] J. Schiller; Mobile Communications; Addison-Wesley, 2003; Chapter 1 - Introduction → History of Mobile Communications

[1.2] R. Read, Nachrichten- und Informationstechnik; Pearson Studium 2004; Easy-to-read introduction to communications and information technology

[1.3] K. Beyerer, B. Mathis (eds.); So weit das Auge reicht - Die Geschichte der optischen Telegraphie; Karlsruhe, Braun, 1995, ISBN 3-7650-8150; Popular science book on the history of optical telegraphy

[1.4] T. S. Rappaport, Wireless Communications, Prentice Hall, 1999

References and further reading

[1.5] www.handywellen.de

[1.6] www.umtslink.at/mobilfunkgeschichte.htm

[1.7] http://www.spiegel.de/netzwelt/technologie/0,1518,405336,00.html

[1.8] http://www.tisa.org/technologies/tmc/

[1.9] http://www.alte-telefone.de/seite50.htm

[1.10] http://www.mbraetz.de/krypto_1402

[1.11] www.wikipedia.de (especially images)

[1.13] http://epp.eurostat.ec.europa.eu/

[1.14] http://www.t-mobile.de/hotspot/1,9646,13391-_,00.html?svl=100